Industrial Vision

Lecture 14 - Two-View Stereo & Epipolar Geometry

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Where we are

We can **calibrate** a camera, and given a **known** set of points in three dimensions we can say where the camera stands, by PnP. Both assumed a world that somebody had already measured, and the map has to come from somewhere.

Today two cameras are placed in the world with no map at all. Recall that a **pure rotation** conceals depth, since $H = KRK^{-1}$ and no parallax arises; as soon as the camera **translates**, points shift by an amount depending on their depth, and it is that parallax we use to establish correspondence.

Today we consider two cameras, either a stereo pair or a single camera that has been translated. A world point $\mathbf{X}$ is imaged at $\mathbf{x}$ in the left view and at $\mathbf{x}'$ in the right view. Two things are required in order to obtain depth.

1. The correspondence search, in which the point $\mathbf{x}'$ matching $\mathbf{x}$ is located.

2. Triangulation, in which both rays are back-projected and their intersection in three dimensions gives $\mathbf{X}$.

The principal result of this lecture is that the correspondence problem is **not** a two-dimensional search. The geometry constrains $\mathbf{x}'$ to lie on a **line**, so that the search is one-dimensional.

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The stereo setup

There are two optical centers, C on the left and C' on the right, and the line CC' joining them is the **baseline**. We take the **left camera as the world frame**, so that C' is located at some pose $(R, \mathbf{t})$.

- In **dense stereo** a depth is computed for every pixel, giving a **depth map** whose dimensions match those of the input.
- In **sparse stereo** depths are computed only at matched feature points, after which the result may be extended by region growing.

There are two configurations, converging and parallel.

- In the **parallel**, or rectified, configuration $R = I$ and the translation is along the image x-axis only, so that $\mathbf{t} = (T_x, 0, 0)$. This is the simplest case, and the epipolar lines become horizontal scanlines.
- In the **converging** configuration the pose $(R, \mathbf{t})$ is general. This is more realistic but harder to arrange exactly.

We derive the general converging case, from which the parallel case follows as a special case.

- A forward translation T_z, which resembles a zoom, does not give parallel stereo, since the baseline must lie in the image plane.

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Disparity carries depth

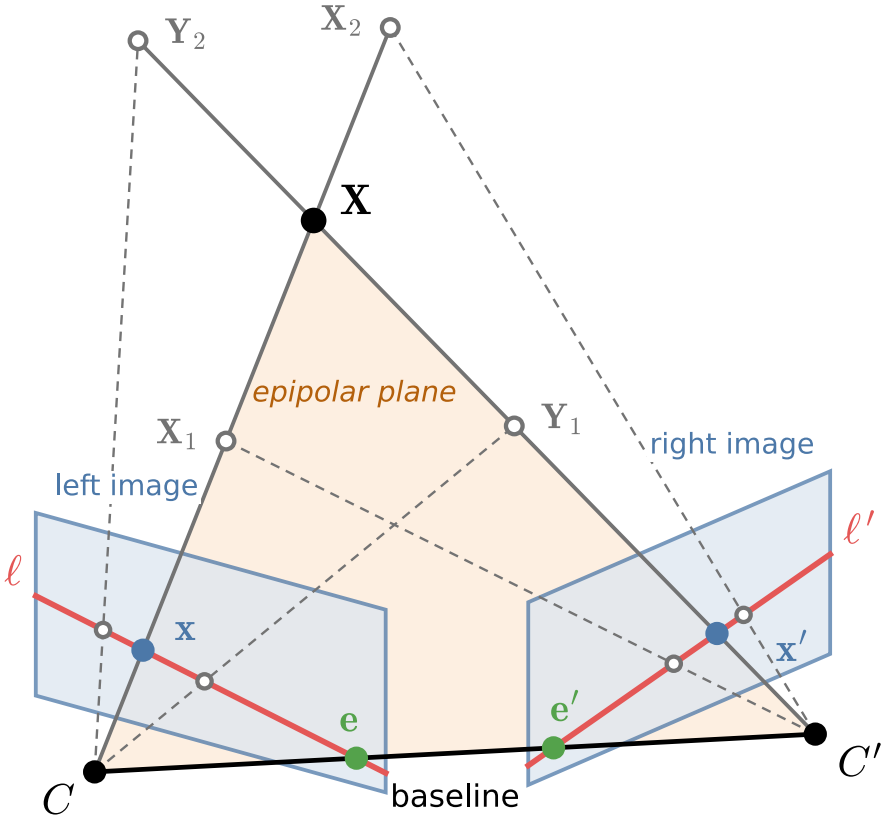
Why does a translation reveal depth? The projection equation $x = fX/Z$ already provides the answer. For a translated camera the shift in the image between $\mathbf{x}$ and $\mathbf{x}'$, which is called the **disparity**, satisfies

$$\text{disparity} \;\propto\; \frac{1}{Z}.$$

- **Near** points move a great deal between the two views, whereas **distant** points move very little. A distant mountain barely shifts, whereas a finger held at arm's length shifts markedly.
- Recovering Z therefore amounts to measuring d, which in turn amounts to **finding the correspondence**. Everything depends on the match.

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The two-view geometry at a glance

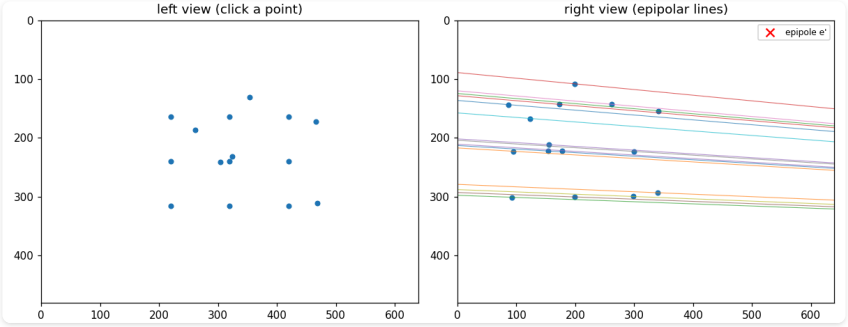


Two cameras, each with a center, C and C', and an image plane.

- A point $\mathbf{x}$ in the left image back-projects to a **ray** from C. The points $\mathbf{X}_1$, $\mathbf{X}$ and $\mathbf{X}_2$ all lie on it, so the left view cannot tell them apart.
- In the right view they land at different places, but all of them on one line: the image of the ray, which is the **epipolar line** $\boldsymbol{\ell}'$.
- Similarly, the ray from C' through $\mathbf{x}'$ images in the left view as the epipolar line $\boldsymbol{\ell}$, which passes through $\mathbf{x}$.
- The **epipoles** are the images of the other center: $\mathbf{e}'$ of C and $\mathbf{e}$ of C'. They lie where the **baseline** CC' pierces the images, and each epipolar line passes through its epipole.
- The baseline and $\mathbf{X}$ span the **epipolar plane**, shaded, which cuts the images in $\boldsymbol{\ell}$ and $\boldsymbol{\ell}'$.

The terminology of epipolar geometry

Epipolar geometry follows from the **coplanarity** of the two camera centers and the scene point.



On the left are the points x. On the right each match lies on its epipolar line l′. The lines are nearly parallel here, because the epipole lies far outside the frame.

- The **epipole** $\mathbf{e}'$ is the image of the other camera's center, that is, the point at which the baseline meets the image plane.
- The **epipolar line** ℓ' is the image, in the right view, of the ray back-projected through $\mathbf{x}$. The match $\mathbf{x}'$ **must** lie on it.
- The **epipolar plane** is the plane through $\mathbf{X}$, C and C'. It contains both centers and both epipoles.
- The **epipolar pencil** is the family of epipolar lines obtained by rotating that plane about the baseline. Every line of the pencil passes through $\mathbf{e}'$.

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An algebraic expression for the epipoles

The epipole is simply the **projection of the other camera center**:

$$\mathbf{e}' = P_R \tilde{C}, \qquad \mathbf{e} = P_L \tilde{C}'.$$

The point $\mathbf{e}'$ is where the center of camera L appears in image R. **Every** epipolar line in the right image passes through $\mathbf{e}'$, because every scene ray also passes through C, so that its image in the right view must contain the image of C. This common point of intersection is the vertex of the pencil.

- The entire construction depends **only** on the two cameras, through the intrinsic matrices (K_L, K_R) and the relative pose $(R, \mathbf{t})$, and **not on the scene**.
- The question of where to search is therefore settled once the pair of cameras has been calibrated, whatever scene they are directed at.

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How the epipolar line is constructed

We can now state precisely how the epipolar line in the right view is obtained for a point $\mathbf{x}$ in the left view.

1. **Back-project** a ray from C through $\mathbf{x}$.
2. Choose **two points** on that ray and **project** both into the right image using P_R.
3. The **line through those two image points**, obtained as a cross product, is the epipolar line ℓ' in the right view.
4. Any true match $\mathbf{x}'$ satisfies $\ell'^{\top} \tilde{\mathbf{x}}' = 0$.

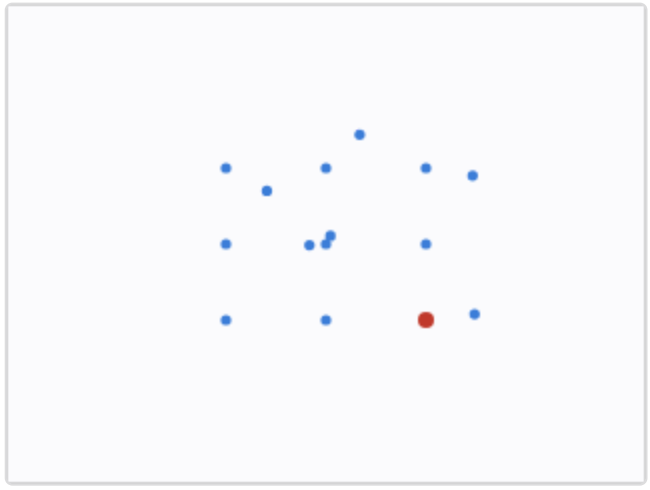
When carried out algebraically, steps 1 to 3 reduce to a single 3×3 matrix F satisfying $\ell' = F\,\tilde{\mathbf{x}}$. This is the **fundamental matrix**, and its derivation is the subject of the next lecture. We first examine the geometry.

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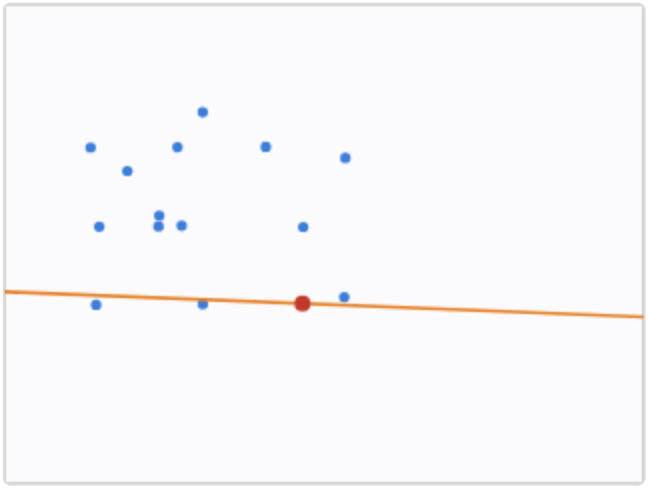
Inline demonstration

Move the cursor over the **left** view. Its **right epipolar line** $l' = Fx$ is drawn live — the true match (red) always sits on it.

left view — x



right view — $l' = Fx$



`nearest catalogued match: residual |x′ᵀ F x| =` **`0.0e+0`** `✓ the match lies on the epipolar line (1-D search)`

As the cursor moves over the left view, the **epipolar line** $l' = Fx$ is drawn in the right view, and the true match always lies on it, so that a two-dimensional search becomes a **one-dimensional** one. The same calibrated scene is used by the Python checks, which may be run with

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Summary

- **Stereo consists of correspondence together with triangulation.** There are two camera centers joined by a **baseline**, and the left camera is taken as the world frame so that C' lies at $(R, \mathbf{t})$.
- The **disparity is proportional to** $1/Z$, so that measuring the displacement of a match recovers the depth. For a parallel configuration $d = fb/Z$. Dense stereo gives a depth map and sparse stereo gives depths at feature points.
- In **epipolar geometry** the epipole is $\mathbf{e}' = P_R \tilde{C}$, the match must lie on the epipolar line, the epipolar plane contains both centers, and the **pencil** consists of the lines through $\mathbf{e}'$. The geometry depends on the **cameras alone** and not on the scene.
- The homogeneous representation in two dimensions gives incidence as $\boldsymbol{\ell}^\top \tilde{\mathbf{x}} = 0$, the line through two points as $\boldsymbol{\ell} = \mathbf{p} \times \mathbf{q}$, and the intersection of two lines as $\tilde{\mathbf{x}} = \boldsymbol{\ell} \times \mathbf{m}$.
- The correspondence search is thereby reduced from **two dimensions to one**, which is the principal result of the lecture.

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References

1. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge Univ. Press, 2004. See **Ch. 9, "Epipolar Geometry and the Fundamental Matrix"**, in particular §9.1 on epipolar geometry and §9.2 on the pencil and the epipoles.
2. D. A. Forsyth and J. Ponce, *Computer Vision: A Modern Approach*, 2nd ed. Pearson, 2012. See Ch. 7, "Stereopsis", on disparity and correspondence.

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